



Lecture 10: TypeScript Integration

INTERVIEW QUESTION

How do you enable and configure TypeScript natively inside a Svelte component?

TypeScript has become the industry standard for writing robust, maintainable web applications. Fortunately, modern Svelte projects (such as those generated by SvelteKit or Vite) come with native TypeScript support fully configured out of the box.

To enable TypeScript within a specific `.svelte` component, you do not need to install complex loaders or write custom Webpack configurations. You simply add a language attribute to your script tag.

The `lang="ts"` Attribute

By adding `lang="ts"` to the opening `<script>` tag, you instruct the Svelte compiler's preprocessor to expect TypeScript syntax instead of standard JavaScript.

The screenshot shows a code editor window titled "Calculator.svelte". The code is as follows:

```
1 <script lang="ts">
2   let total: number = $state(0);
3
4   function add(a: number, b: number): number {
5     return a + b;
6   }
7 </script>
8
9 <button onclick={() => total = add(total, 1)}>
10   Total: {total}
11 </button>
```

Annotations in the image:

- A callout box pointing to the `lang="ts"` attribute on line 1 says: "Simply add this attribute. The compiler now safely processes TS syntax".
- A callout box pointing to the type annotations on line 4 says: "Types enforce safety in your editor".
- A callout box pointing to the Svelte markup on line 11 says: "Standard markup remains unchanged".

When the Svelte compiler encounters a file with `lang="ts"`, a preprocessor intercepts the `<script>` block. Because browsers cannot execute TypeScript natively, the preprocessor strips away all the type annotations (`: number`, interfaces, generics) and converts the block back into plain, standard JavaScript. After the types are stripped, the standard Svelte compiler takes over to generate the reactive framework code.

KEY TAKEAWAY

Enabling TypeScript in Svelte is as simple as writing `<script lang="ts">` . The underlying preprocessor automatically strips the types during the build step, ensuring that only standard JavaScript ships to the browser.

INTERVIEW TIP

To impress the interviewer: It is vital to clarify what the compiler actually does. Many junior developers incorrectly assume that compiling a TypeScript Svelte component means the framework performs a massive, strict type-checking pass that slows down the build. In reality, Vite and Svelte simply *strip* the types (using fast tools like ESBuild). The actual deep type-checking is deferred to your IDE (like VS Code with the Svelte extension) or run as a separate continuous integration task via the `svelte-check` CLI tool.

Summary

TypeScript is enabled at the component level by adding the `lang="ts"` attribute to the script tag. This tells the build tools to process the block as TypeScript, allowing you to use static typing in your editor while the compiler safely strips the annotations away before generating the final browser-ready JavaScript bundle.

	STANDARD SCRIPT (<code><script></code>)	TYPESCRIPT SCRIPT (<code><script lang="ts"></code>)
Language Syntax	Vanilla JavaScript only	Full TypeScript support
Compiler Action	Directly compiles to reactive JS	STRIPS types, then compiles to reactive JS
Type Safety	None	Enforced by IDE and <code>svelte-check</code>